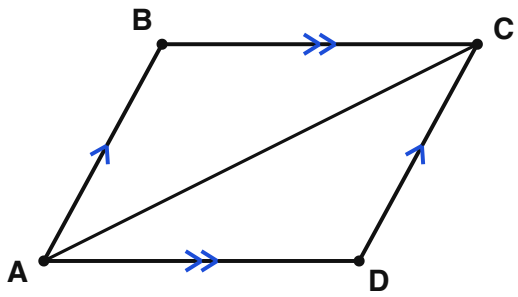
Proof Practice: Answer Key

Parallelograms, rhombi, rectangles, squares, kites and converses I diagrams are not drawn to scale

These are model proofs. Yours can differ in order or use a different valid theorem, as long as every step has a correct reason.

1. Diagonal splits a parallelogram ★☆☆

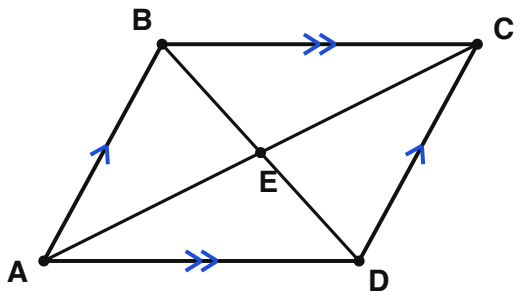


Given: ABCD is a parallelogram with diagonal AC.

Prove: △ABC ≅ △CDA

	Statements	Reasons
1	ABCD is a parallelogram	Given
2	AB ≅ CD and BC ≅ DA	Parallelogram opposite sides thm
3	AC ≅ CA	Reflexive property (shared side)
4	△ABC ≅ △CDA	SSS

2. Diagonals of a parallelogram bisect each other ★★☆

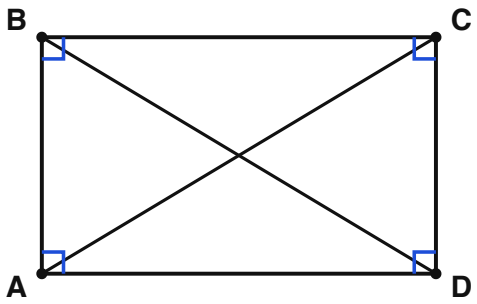


Given: ABCD is a parallelogram. Diagonals AC and BD intersect at E.

Prove: AE ≅ CE and BE ≅ DE

	Statements	Reasons
1	ABCD is a parallelogram	Given
2	AB ∥ DC	Definition of parallelogram
3	∠BAE ≅ ∠DCE and ∠ABE ≅ ∠CDE	Alternate interior ∠'s thm (AC and BD are transversals)
4	AB ≅ CD	Parallelogram opposite sides thm
5	△ABE ≅ △CDE	ASA
6	AE ≅ CE and BE ≅ DE	CPCTC (corresponding parts of congruent triangles are congruent)

3. Diagonals of a rectangle are congruent ★★☆

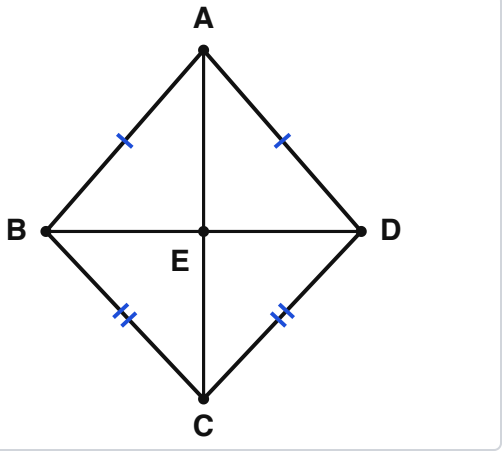


Given: ABCD is a rectangle (four right angles; opposite sides ≅ and ||) with diagonals AC and BD.

Prove: AC ≅ BD

	Statements	Reasons
1	ABCD is a rectangle	Given
2	AB ≅ DC	Opposite sides of a rectangle are ≅ (a rectangle is a parallelogram)
3	∠ABC and ∠DCB are right angles	Definition of rectangle
4	∠ABC ≅ ∠DCB	Right angles congruence thm
5	BC ≅ CB	Reflexive property
6	△ABC ≅ △DCB	SAS
7	AC ≅ DB	CPCTC

4. A kite's diagonal is a perpendicular bisector (uses a converse) ★★☆

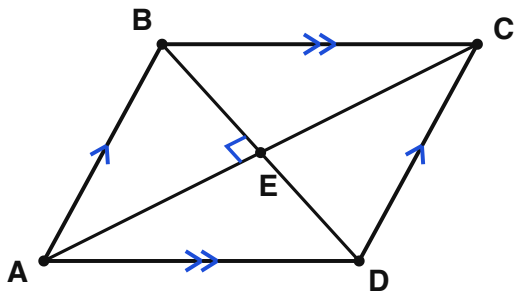


Given: ABCD is a kite with AB ≅ AD and CB ≅ CD. The diagonals meet at E.

Prove: Line AC is the perpendicular bisector of BD.

	Statements	Reasons
1	AB ≅ AD	Given
2	A lies on the perpendicular bisector of BD	Converse of perpendicular bisector thm (A is equally far from B and D)
3	CB ≅ CD	Given
4	C lies on the perpendicular bisector of BD	Converse of perpendicular bisector thm (C is equally far from B and D)
5	Line AC is the perpendicular bisector of BD	Two points determine a line, and A and C are both on the perpendicular bisector

5. Parallelogram with perpendicular diagonals is a rhombus ★★★

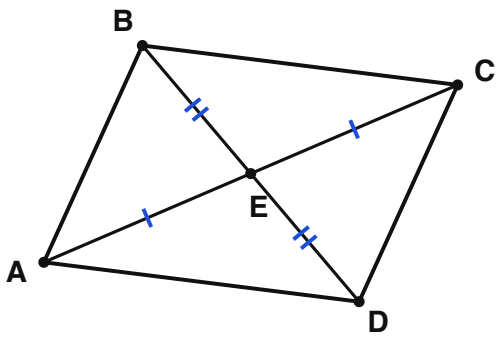


Given: ABCD is a parallelogram. Diagonals AC and BD intersect at E and AC ⊥ BD.

Prove: ABCD is a rhombus.

	Statements	Reasons
1	ABCD is a parallelogram	Given
2	AE ≅ CE (E is the midpoint of AC)	Parallelogram diagonals thm (diagonals bisect each other)
3	AC ⊥ BD	Given
4	BD is the perpendicular bisector of AC	Steps 2 and 3: BD passes through the midpoint of AC at 90°
5	BA ≅ BC	Perpendicular bisector thm (B is on the perpendicular bisector of AC)
6	AB ≅ CD and BC ≅ DA	Parallelogram opposite sides thm
7	AB ≅ BC ≅ CD ≅ DA	Substitution (steps 5 and 6)
8	ABCD is a rhombus	Definition of rhombus (4 ≅ sides)

6. Diagonals that bisect each other make a parallelogram (uses a converse) ★★★

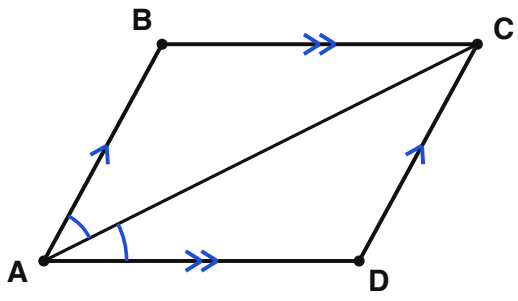


Given: In quadrilateral ABCD, diagonals AC and BD intersect at E with AE ≅ CE and BE ≅ DE.

Prove: ABCD is a parallelogram.

	Statements	Reasons
1	AE ≅ CE and BE ≅ DE	Given
2	∠AEB ≅ ∠CED and ∠AED ≅ ∠CEB	Vertical angles congruence thm
3	△AEB ≅ △CED	SAS (AE ≅ CE, ∠AEB ≅ ∠CED, BE ≅ DE)
4	AB ≅ CD	CPCTC
5	△AED ≅ △CEB	SAS (AE ≅ CE, ∠AED ≅ ∠CEB, DE ≅ BE)
6	AD ≅ CB	CPCTC
7	ABCD is a parallelogram	Converse of parallelogram opposite sides thm (both pairs of opposite sides ≅)

7. Diagonal bisects an angle of a parallelogram: rhombus (uses a converse) ★★★

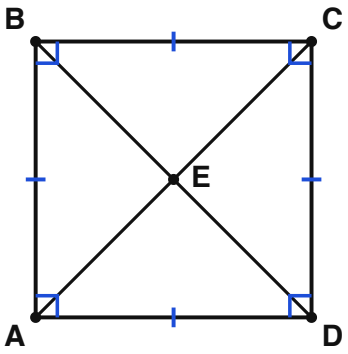


Given: ABCD is a parallelogram. Diagonal AC bisects ∠DAB, so ∠BAC ≅ ∠CAD.

Prove: ABCD is a rhombus.

	Statements	Reasons
1	ABCD is a parallelogram	Given
2	AB ∥ DC	Definition of parallelogram
3	∠BAC ≅ ∠DCA	Alternate interior ∠'s thm (transversal AC)
4	∠BAC ≅ ∠CAD	Given (AC bisects ∠DAB)
5	∠CAD ≅ ∠DCA	Transitive property (steps 3 and 4)
6	DA ≅ DC	Converse of base ∠'s thm (in △ADC, the sides opposite the congruent angles are ≅)
7	AB ≅ CD and BC ≅ DA	Parallelogram opposite sides thm
8	AB ≅ BC ≅ CD ≅ DA	Substitution (steps 6 and 7)
9	ABCD is a rhombus	Definition of rhombus (4 ≅ sides)

8. Challenge: diagonals of a square are congruent and perpendicular ★★★



Given: ABCD is a square with diagonals AC and BD intersecting at E.

Prove: AC ≅ BD and AC ⊥ BD

	Statements	Reasons
1	ABCD is a square	Given
2	AB ≅ BC ≅ CD ≅ DA, and all four angles are right angles	Definition of square
3	∠ABC ≅ ∠DCB	Right angles congruence thm
4	△ABC ≅ △DCB	SAS (AB ≅ DC, ∠ABC ≅ ∠DCB, BC ≅ CB)
5	AC ≅ DB	CPCTC
6	AB ≅ AD and CB ≅ CD	Step 2
7	A and C both lie on the perpendicular bisector of BD	Converse of perpendicular bisector thm
8	Line AC is the perpendicular bisector of BD, so AC ⊥ BD	Two points determine a line